Yiyi Cai

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Education

2026– **Stanford University**, Stanford, CA
Doctor of Philosophy, Computer Science

2025-26 **University of Cambridge**, Cambridge, UK
Master of Philosophy, Advanced Computer Science, with distinction

2021-25 **California Institute of Technology**, Pasadena, CA
Bachelor of Science, Electrical Engineering, GPA 4.1/4.3

Papers (* indicates alphabetical ordering)

- **Yiyi Cai***, Yongtao Zhan*, Alexander Zlokapa*. Fast mixing of the SYK model at high temperatures. arXiv:2610.05953.
- Jinge Bao*, **Yiyi Cai***, Zhong-Xia Shang*. Optimal tolerant testing of Lindbladian dissipation. arXiv:2609.38160.
- Akshar Ramkumar, **Yiyi Cai**, Yu Tong and Jiaqing Jiang. High-temperature fermionic Gibbs states are mixtures of Gaussian States. *Physics Review Letters*, vol. 137, no. 2, p.020601, 2026. **QIP 2026**.
- **Yiyi Cai**. Optimal detection of dissipation in Lindbladian dynamics. arXiv:2603.17736.
- **Yiyi Cai**. Improving parameters of asymptotically good quantum LDPC codes via stronger product expansion. Senior Thesis (Major).
- **Yiyi Cai**, Yu Tong, and John Preskill. Stochastic error cancellation in analog quantum simulation. *In 19th Conference on the Theory of Quantum Computation, Communication and Cryptography*. **Outstanding Paper Prize at TQC 2024.**
- Elena Scire, Lee Bennett, Judy Adler, Sergio Fajardo-Acosta, Robert Fogg, Elise Furlan, Jacob Llamas, Peter Bahariance, **Yiyi Cai**, Trinity Chung, Garni Gharibian, Emily Hu, Julianna Jin, Matteo Kimura, Aaron Lee, Michael Mansour, Mahak Mathur, Andy Sun, Jasmine Terrones, Jingchao Zhong, and Bethany Ehlmann. Lunar trailblazer ground system development. *Proceeding SPIE 13098, Observatory Operations: Strategies, Processes, and Systems X*, 2024.

Awards & Honors

2026 National Science Foundation Graduate Student Fellowship
2026 Stanford Graduate Fellowship
2025 Gates Cambridge Scholarship
2025 National Defense Science and Engineering Graduate Fellowship (declined)
2024 Outstanding Paper Prize at 2024 TQC

2023	Mellon Mays Undergraduate Fellowship
2023	Arthur R. Adams Summer Undergraduate Research Fellowship
2022	Doris Everhart Quantum Summer Undergraduate Research Fellowship

Research Experience

2025 - 2026	Student Researcher, University of Cambridge
2023 - 2025	Student Researcher, Institute of Quantum Information and Matter, California Institute of Technology
2024	Quantum Computing Research Analyst, Global Technology Applied Research, JP-Morgan Chase & Co.
2022	Student Researcher, Quantum Machine Learning for High Energy Physics, California Institute of Technology
2021 - 2023	Missions Operations & Ground Data Science Intern, Lunar Trailblazer Mission, NASA & California Institute of Technology

Talks & Seminars

- ◇ University of Waterloo Institute for Quantum Computing Math and CS seminar; June 2026
- ◇ Contributed talk at QIP; Jan. 2026
- ◇ Academia Sinica seminar; Aug. 2025
- ◇ Contributed talk at TQC; Sep. 2024
- ◇ Caltech Summer Undergraduate Research Fellowship Seminar; Aug. 2023
- ◇ Caltech Summer Undergraduate Research Fellowship Seminar; Oct. 2022

Teaching

The Coding School

Teaching Assistant & Curriculum Developer

Summer 2025	Quantum Summer Institute

California Institute of Technology

Teaching Assistant

Spring 2025	IDS 157 Statistical Inference
Fall 2025	EE 111 Signal-Processing Systems and Transforms
Spring 2024	EE 10b Embedded Systems
Winter 2024	EE 10a Digital Logic
Fall 2024	EE 55 Mathematics of Electrical Engineering

Academic Service

Journal Reviews

PRX, PRX Quantum

Conference Reviews

QIP 2027, FOCS 2026, ICALP 2026, ITCS 2026, QIP 2026

Last updated: October 19, 2026